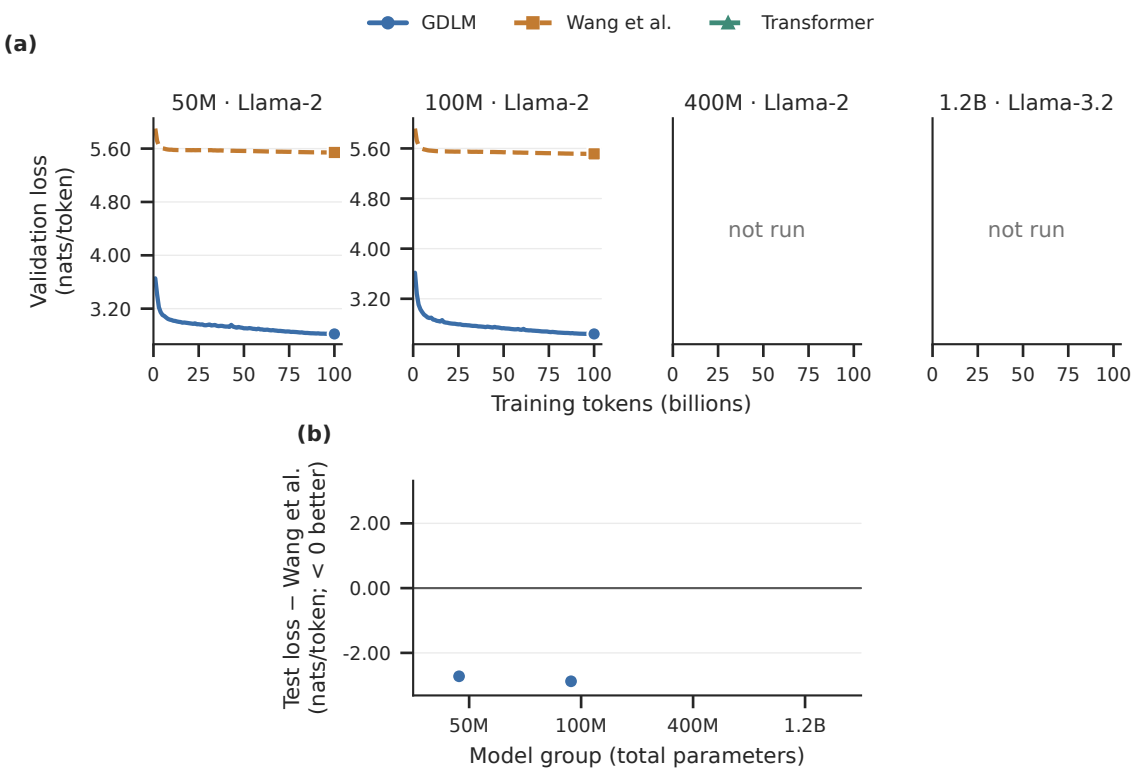
(a)
GDLM
Wang et al.
Transformer
50M · Llama-2
100M · Llama-2
400M · Llama-2
1.2B · Llama-3.2
Validation loss (nats/token)
5.60
4.80
4.00
3.20
not run
not run
0
25
50
75
100
Training tokens (billions)
(b)
Test loss − Wang et al. (nats/token; < 0 better)
2.00
0.00
-2.00
50M
100M
400M
1.2B
Model group (total parameters)